

# Opportunities and Challenges Related to AI in Education

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**Abstract**—Artificial Intelligence (AI) in education is a structural shift from traditional learnings, that carries both risks and potential to transform individuals, communities or a nation. This paper examines the dual nature of AI's influence on education. It examines the main advantages, such as the ability to provide guidance to individuals at a large scale, increasing the productivity through automation, and promote greater accessibility for a range of learners. At the same time, it tackles the crucial problem that need to be overcome for responsible execution and implementation of AI. These include concerns about data privacy and security, the possibility of academic dishonesty and the dehumanization of learning, the threat of algorithmic bias sustaining social bias and prejudice, and the danger of the digital divide growing the gap of poor and rich further. The paper makes the case that a purposeful, human-centric framework is necessary to both realize AI's potential and mitigate its risks.

**Keywords**—Artificial Intelligence, AI in Education, Personalized Learning, Algorithmic Bias, Digital Divide, Educational Technology, Data Privacy.

## I. INTRODUCTION

The term “Artificial Intelligence” was first created by John McCarthy at the 1956 Dartmouth Conference, but the core ideas and concepts of AI can be traced back much farther in history, such as in the field science and philosophy of ancient Greece, long before modern pioneers like Alan Turing shaped the AI technology.[1].

The capability of computers to have a thought process and behavioural patterns exactly like that of humans, is called artificial intelligence (AI), and also, those efforts that are made to build such computerised systems that can imitate human thinking and behaviour. Given its potential to be a potent element in guaranteeing economic development, artificial intelligence is a perfect fit to be set as the fundamental element for the 5<sup>th</sup> Industrial Revolution [2].

The field of teaching and learning is being drastically altered by artificial intelligence (AI). Artificial Intelligence (AI) in education refers to the use of sophisticated computer systems to echo the human intelligence and implement them in areas such as learning, solving a problem, and making decisions, thereby completing tasks that have historically

required human cognition. This entails utilizing technology to address the complex social, emotional, academic, and cognitive elements that impact students' learning.

AI integration's main objective is to radically improve and streamline learning experiences rather than just bring new technology into the classroom. By enhancing the capacities of both teachers and students, a more effective, individualised, and efficient educational ecosystem is to be created.

The central argument is that a balanced, human-centric framework is essential to ensure AI is leveraged as a supportive tool for educators, an empowering resource for students, and a catalyst for equitable learning opportunities for all.

The ability to overcome challenging situations is typically seen as intelligence. AI focusses on the methods to complete tasks, when a complex dataset is given. Whether that problem is being solved by a computer program or a human or any other creature, the means to solve the problem, that must be used are all linked to the problem itself, that the situation presents [1].

AI education began in university computer science programs, which demanded sophisticated programming skills that were too high for students in grades K–12. Using syntax-based programming to scaffold K–12 students' understanding of AI principles presented difficulties for educators [3].

## II. TRANSFORMING EDUCATIONAL OUTCOMES

The strong goal of creating more individualised, effective, and equitable learning environments is driving the movement to incorporate AI into education. Proponents view AI as a revolutionary force that can tackle most of the difficulties that occur in the field education, such as teacher burnout and student apathy, rather than as a simple incremental update.

### A. Personalized Learning at Scale

Students who utilize the AI-powered personalised learning tools scored significantly higher grades than those students who opt for the traditional learning methods. Their performance was also more consistent; this indicates that AI-powered learning tools helps to create a more stable learning path [4].

AI's potential to eliminate the conventional approach to education and substitute it with highly customised learning experiences is its greatest use. Artificial intelligence (AI) powered learning systems evaluate a student's performance and learning patterns in real time to modify the level of syllabus, provide resources to overcome knowledge gaps, and offer focused assistance. This increases confidence and participation of students by letting them grow and improve at their own capacity.

AI powered tutoring systems such as MATHia, serve as virtual one-on-one tutors that offer detailed instructions and real-time, personalised feedback. Students can clear up misunderstandings and make iterative improvements to their work without the delays that are typical in traditional settings thanks to this instantaneous feedback loop, which is essential for the learning process.

AI-powered personalised learning paths are positively correlated with increased retention, engagement, and academic success. The study clarifies the subtle impacts across various academic levels and disciplines by identifying critical success criteria and potential pitfalls in the application of personalised learning [5].

Based on IBM's Watson platform, Georgia Institute of Technology deployed "Jill Watson," which is an AI-powered teaching assistant. Jill, which was implemented in an online Masters in Computer Science course, was made to automatically address frequently asked issues by students in the online forum. To appropriately respond to frequently asked questions by students, an AI assistant was trained on a dataset of more than 40,000 forum postings from earlier parts of the course.

AI-powered program was made and implemented in the University of Murcia in Spain, a chatbot called Lola that assists students with inquiries about the campus events and academic activities. The chatbot sowed a high level of efficiency, answering over 38,708 questions correctly more than 91% of the time.

An AI algorithm was created by a Stanford researcher to help students who are having trouble with self-paced digital learning. A machine-learning program that could anticipate when a learner would become stuck and begin "wheel-spinning" and then suggest a pertinent answer was tested in the study. 1,170 Ugandan schoolchildren who had utilised tablets to master English reading abilities through movies and mini-games provided performance statistics for the program's training. Even before a youngster started a new lesson, the computer could anticipate if they would engage in "wheel-spinning". The recommendations made by the model and the human expert were identical in four of the six circumstances. According to the study's findings, AI can be used to recognise student issues and facilitate the assistance of a large number of students by a small number of human teachers [6].

According to research by the adaptive learning firm "Knewton", students who used their AI-powered program saw a 62% increase in test scores when compared to those who did not. Students received individualised feedback and

guidance from the program, which customised the learning process to meet their unique requirements and skill levels [7].

## B. Enhancing Teacher Efficiency

Jill Watson greatly reduced the time it took to respond to student enquiries and lightened the workload of human teaching assistants, allowing them to concentrate on more complicated student requirements.

AI technologies can help teachers design dynamic classrooms that enhance learning and personalise it. Furthermore, by helping with grading and coming up with ideas for new assignments, AI tools can drastically cut down on the amount of time spent on some instructional tasks while freeing up time for engineering instructors to concentrate on crucial instructional tasks like leading class discussions, mentoring students, and keeping an eye on their teamwork [8].

Students mostly utilise AI tools for information searching, comprehending sentences or words of other languages, and producing documents that are grammatically accurate. [9].

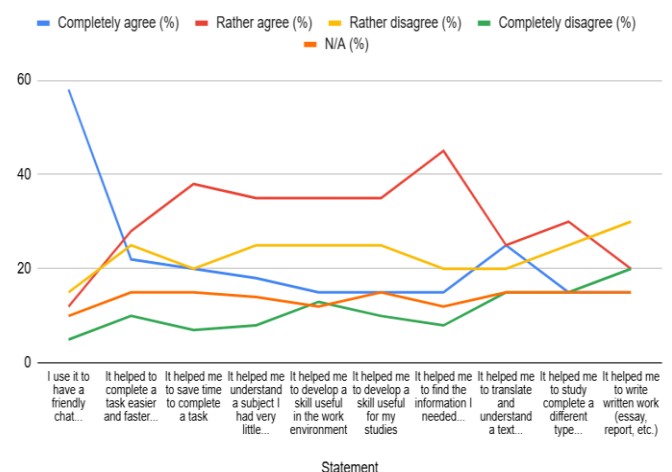


Fig. 1. Statistics of perceived benefits of using ChatGPT, according to various users [9].

According to 75% of respondents, the most significant advantage of AI tools is that they facilitate learning for students who face language hurdles, and 70% concur that they increase access to global knowledge. According to 65% of respondents, the third most valuable advantage of AI technologies is that they facilitate learning for students with disabilities [9].

In recent years, instructors have been able to enhance the learning experience for younger students because to the development of more age-appropriate technology and software. Everyone has the opportunity to learn about and apply artificial intelligence (AI) in their daily lives thanks to the abundance of technology available to them, such as chatbots and translation apps. This makes it possible for educators to use the AI technologies that are already available to instill AI literacy in young students [3].

In recent years, fields such as artificial intelligence, machine learning and deep learning have experienced a significant revival. Moreover, there is a growing belief among analysts

that AI is set to have a profound impact on every aspect of society [10].

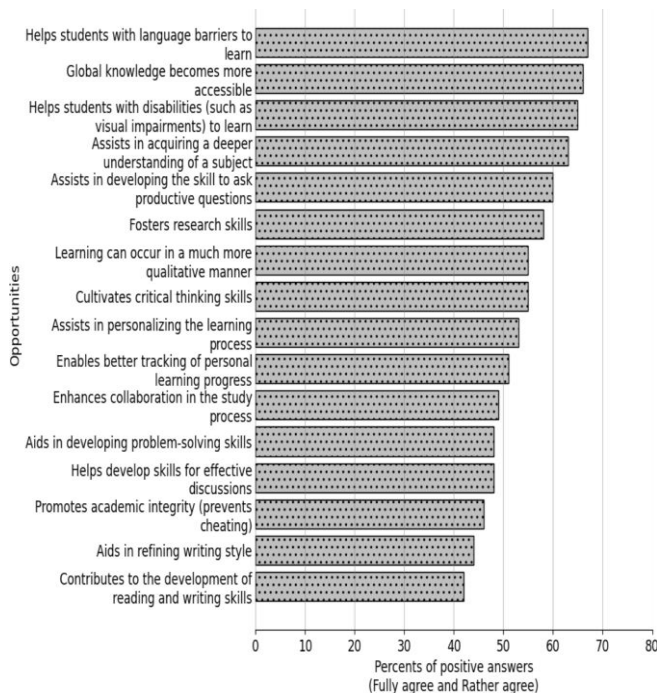


Fig. 2 Frequency of use for various AI tools, as reported by users [9].

This shows that AI has numerous benefits in the field of education both for learners and educators.

### C. Fostering Accessibility for Children with Special Need

AI has a lot of promise to make classrooms more inclusive. AI-powered solutions like text-to-speech and speech-to-text can increase the accessibility of educational materials for students with disabilities. Language barriers can be eliminated for multilingual learners by using AI-driven translation to instantly translate lectures and course materials into their native tongues. AI can help democratise access to top-notch educational resources by offering individualised support at scale, levelling the playing field for students from all backgrounds.

An artificial intelligence platform that offers adaptive learning for special needs, after determining their mathematical strengths, students can pose more difficult math questions to their peers. On the other hand, it can offer resources to help kids who struggle with language acquire more fundamental language abilities. Additionally, this platform can track pupils' growth over time [11].

The University of Alicante aimed to enhance learning opportunities and accessibility for visually impaired students, who frequently encounter considerable obstacles when attempting to navigate campus settings and getting educational resources. The institution created "Help Me See," an AI-powered mobile application, that helps visually challenged students by utilising computer vision library and machine learning algorithm. Through the recognition and narration of objects, words, and other environmental factors, the software facilitates their autonomous and efficient navigation of the campus. For visually challenged students, "Help Me See" has significantly improved campus accessibility, enabling them to navigate academic

environments with more independence and resulting in boosting confidence. As a result, student involvement and engagement in campus events have increased.

AI platforms specifically made for children with special needs can adapt to their strengths and weaknesses giving a high output and improving the level of those children.

## III. CRITICAL CHALLENGES RELATED TO AI

Even though AI has a lot of potential advantages, there are also a lot of serious ethical, social, and practical issues. The positive potential of AI could be undermined and real harm could result if these risks are not addressed.

### A. Algorithmic Bias and Challenges for Educators

Bias is defined as a systematic error in a decision-making process that yields unjust and harmful consequences. The sources of bias in Artificial Intelligence are multifaceted, arising from factors like data acquisition or algorithmic architecture. Machine learning models, have the capability to inculcate and then reproduce the prejudiced patterns from training data, thereby generating biased outputs. Consequently, the identification and mitigation of bias in AI systems and models are imperative to guarantee fairness and equity for all users [12].

Algorithmic bias is one of AI's most pernicious hazards. When AI systems are trained on data that reflects societal biases, they will unavoidably pick up on, reinforce, and frequently magnify those biases on a large scale. This can show up in education in a number of detrimental ways.

There is a greater chance of negative biases as generative AI (GenAI) systems become more prevalent. A startling example of GenAI bias was documented, in which the outputs of models that convert prompts into images such as Midjourney, Stable Diffusion and DALL-E displayed stereotype and racial prejudices. These models exhibited gender bias as they tended to provide images of men when asked to create images of CEOs. The fact that there is a lack of CEOs being women in reality, is reflected in this bias. Additionally, the models produce significantly more images of persons of colour when asked to create images of terrorists or criminals [12].

Since generative AI models can create content that propagates harmful racial biases and regressive gender stereotypes, UNESCO has issued warnings about this threat. It emphasises on how these prejudice and biases can seriously harm the marginalised groups all over the world, damaging the fundamental educational values of equality and equity. Because education is meant for everyone.

Along with concerns about AI enabling plagiarism and compromising academic integrity, privacy risks such as data misuse and unauthorised access, are significant concerns. Teachers question the accuracy and authenticity of data produced by AI, pointing to possible biases and mistakes [13].

Teachers are now facing a difficulty in adapting to the present teaching methodologies and tests to enhance students' cognitive, creative, and critical thinking abilities, despite the fact that AI applications can be very helpful in preparing students curriculum [13].

AI can severely damage the students' abilities of thinking critically and creatively because students have begun to rely on AI for everything. This over-reliance on AI may result in lower educational quality and a lack of participation [13].

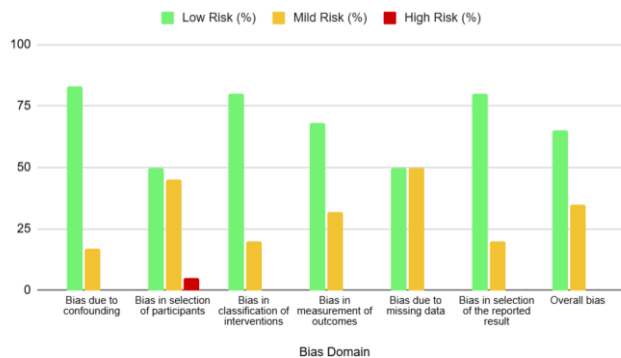


Fig. 3. Distribution of Risk factor on the basis of Bias domains [21].

### B. Data Privacy and Security

The ability of AI to collect and go through the huge amounts of student data, such as academic records, behavioural patterns, and other personally related information quickly, is what makes it effective. Due to this extensive data collection, there are a lot of privacy and security related risks arising, such as the possibility of data misuse for profit by the owners and unauthorised access through data breaches by some hackers.

Securing data at every stage of its lifecycle, from the time it enters AI models to its eventual application in AI systems, requires the establishment of strong data control centres. These controls include data cleaning, encryption, redaction, conversion, and inline classification. It should be made sure that AI models only process and use the authorised and required data by coordinating these procedures with data policies and user consents.

When it comes to privacy and security concerns, the use AI can be very crucial. It can assist in detecting and averting possible flaws and threats to AI systems, such as data poisoning or hacker attacks. By examining patterns and behaviours, AI systems are able to detect anomalies in real time, providing quick responses to stop any attacks [14].

Even with the progress AI has made, there are still many issues remaining, about what can happen if this potent technology ends up in the wrong hands. In order to guarantee that AI is used for the good of humanity and does not present needless risks or harm, it highlights the significance of ethical considerations, strong security measures, and responsible AI development [15].

Increasing use of AI poses one more threat of widespread surveillance, which can have a terrible effect, that diminishes student integrity and reduces the trust that is essential to a positive learning environment.

### C. The Digital Divide by AI

Even though AI is seen as a tool to create and promote equality and accessibility for all, there is a serious risk that its application will further divide the rich from the poor. For underfunded poor schools, especially in the low-income community areas, the high cost of advanced AI platforms becomes unaffordable. It gets worse with the lack of proper

technical gadgets and proper internet connectivity in those areas, The degree of knowledge and ability needed to use AI systems properly is another problem of the digital divide associated with AI. Programming languages, data analysis, and machine learning are just a few of the specialised talents needed to use AI's complicated technology. A skill gap and a wider digital divide may stem from the fact that not all people or communities have access to a good quality education and training in these fields. In the AI-driven economies, people without the training and education may be left far behind, with fewer employment options and less access to AI-powered services [16].

An educational system where schools having proper resources gain from use of AI while schools with less or low-quality resources are the ones that fall further behind, and this is a serious threat of the digital divide due to AI. As students utilising AI tools in education score better than the ones that do not use it.

Artificial intelligence (AI) has the ability to boost economy and open up new business opportunities. However, certain people or business owners with limited financial means may find the cost of creating, deploying, and maintaining AI systems to be very high and unaffordable. The digital divide may be made worse by this economic inequality since people who cannot afford to invest in AI technology may not take advantage of the opportunities they present and as a result they may lack far behind [16].

On the other side, it has been demonstrated that those with greater social standing use digital technology in ways that greatly expand their resources in the areas of education, business, economy, and health [17].

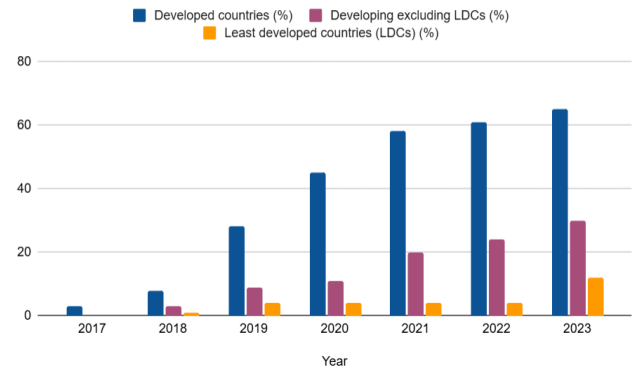


Fig. 4. Comparative trends in AI adoption across developed, developing, and least developed countries (2017–2023).

According to the World Bank, this poses a serious risk to educational equity worldwide and calls for focused investments to guarantee that the advantages of AI are shared among all.

In the MENAAP region i.e. Middle-East, North Africa, Afghanistan and Pakistan, use of AI is very limited, although businesses in these regions are innovating by developing new goods and streamlining procedures. However, very few people or businesses really use AI. Just 26 of the 6,866 businesses were found to be actual AI adopters. When it comes to the use of AI, even wealthier nations like Saudi Arabia and Malta resemble lower-income nations when AI adoption is considered. The only countries with comparatively higher adoption rates are Israel and Bahrain. Overall, the

region shows broad innovation but shallow digital capability, leading to minimal AI use [18].

To overcome the digital divide problem, the main hurdle that comes is of "High Quality resources" everyone cannot afford these High-Tech equipment such as Data Centres, Graphics Processing Units, Wi-Fi etc., so in order to make AI accessible and affordable to each and every individual, developers, innovators and governments all across the globe must focus on developing suitable AI tools that can run smoothly on low-specs resources, and without internet i.e. offline, lightweight AI tools. Such as Gemini Nano, Qwen, GPT-4o mini etc. Big Tech firms like Oracle and Nvidia, are creating their own learning platforms and on that they are providing valuable AI learning material, these courses have everything needed to understand and use AI, and all of this is free of cost, these platforms must be included in schools, colleges and every educational institution to narrow this gap.

Over 1,000 women and young people have already received AI training from UNESCO, along with judicial actors and civil personnel. Continuing this momentum, it will increase efforts to train 5,000 judicial actors and 15,000 civil personnel in 2026–2027. The African AI Talent Incubator, which will be established in Nigeria, South Africa, Rwanda, and Egypt, could provide 1,500 young professionals and researchers with cutting-edge, practical AI capabilities in order to further enhance local capability. Additionally, in order to foster the next generation of AI innovators on the continent, the Africa-Asia Youth Coding Initiative will grow to include 2,000 instructors and students throughout Africa. In partnership with the Knowledge and Innovation Exchange Africa 21 Hub of the Global Partnership for Education, UNESCO's AI Competency Framework will be tailored to African educational systems [19].

UN has Africa as its global priority on ethical use of AI. This Initiative is a call for African innovators, policymakers, as well as international and African investors. This will help to achieve the visionary objectives of the African Union's Agenda 2063 as AI can create an inclusive and ethical growth serving as a powerful tool [19].

#### *D. Academic Integrity in the Age of AI*

Significant queries concerning the nature of learning itself are brought up by the incorporation of AI. Since strong generative AI tools make it simpler for students to cheat or plagiarise, academic misconduct is a major worry for educators.

A review of AI-generated answers for academic tasks revealed that, while the content was predominantly unique, it was flawed by improper citations and a lack of authentic human viewpoints, which are elements that AI is generally incapable of producing [20].

Since most plagiarism checkers are unable to check for AI-generated text, it becomes difficult to point that whether the work of an author is original or not [20].

Due to the constant improvements in technology, communication has become faster and more flexible. It definitely enhances the student learning experience but with it also arises a culture that can result in violating academic integrity. Students can easily find and use the help of contract writers online. Additionally, there are a lot of AI models that generate materials that can be found online [21].

Artificial Intelligence with its strong computation power of completing near impossible tasks within seconds, pose a great threat to academic legitimacy and to all the parties involved in the field of academics.

The concern that an over-reliance on AI may result in the decline of core cognitive abilities like analytical writing, problem-solving, and critical thinking is closely related. AI may unintentionally impede the very advancement it is intended to promote if it disrupts the intellectual competition process.

Guidelines for using GenAI tools were publicly available on 41 (82%) of the 50 university websites that were searched. However, the websites of nine universities did not contain any recommendations. Two of those nine said they were creating their own standards for successfully integrating GenAI into their educational system and were adhering to UNESCO's guidelines for policymakers on GenAI and education [22].

Additionally, the majority of universities mandate that before utilising GenAI tools in assignments, students obtain permission from their lecturers or supervisors. Approximately two-thirds of the guidelines included information on the fields in which GenAI products can be useful. Only with appropriate human oversight and verification of all generated outputs can these technologies help with literature searches and reviews, citation management, summarising, analysing, and draughting papers [22].

Lastly, there is a chance that education will become less human. An overemphasis on automated systems may result in a sterile and alienating educational experience by reducing opportunities for peer-to-peer engagement and the crucial teacher-student feedback loop.

#### **IV. THE PATH FORWARD: A HUMAN-CENTRIC FRAMEWORK**

It takes a purposeful, human-centred strategy to navigate the complexities of AI in education, putting teachers and their interactions with students at the heart of the change.

The response to AI revealed a gap between how higher education has adjusted to it and how quickly these AI capabilities are developing. Higher education has had to adjust to the spread and integration of AI technologies in the classroom as the technology continues to make significant strides. These reactions do have some merit, though, and while examining the gap that has begun to grow, it is important to take into account their background and motivations [23].

##### *A. The Evolving Role of the Educator*

The traditional role of the teacher as a "content deliverer" is becoming less and less relevant as AI tools proliferate. The value of the educator is changing to that of a "learning architect" or "facilitator" of educational experiences. Under this model, the instructor creates engaging learning spaces, selects resources (such as content produced by artificial intelligence), and leads group projects that are inquiry-based.

The emphasis shifts to developing abilities that AI cannot imitate, such as empathy, creativity, critical thinking, and teamwork. And inculcate those skills into the young minds so that they can think of their own rather than being totally dependent on the use of AI. None of the above-mentioned abilities can be taught by AI, they can only be taught by physical teaching and in presence of a guide.

It is already feasible for the human brain to "crossbreed" with a computer, which will effectively force educators to develop new functions, methodologies and completely different pedagogies for a new learning and teaching environment [24].

The role of educators has now also advanced from just teaching to now monitoring what the AI is delivering also. Teachers and professors must know how to use AI that can catch the AI generated content to maintain academic integrity.

Universities should, in effect, reconsider their roles, teaching approaches and frameworks to cope with AI and their owners. Institutions providing higher education also anticipate the amplitude of opportunities and difficulties that will arise with the integration of AI in learning and teaching. These approaches provide possibilities for achieving universal education while fostering lifelong learning, all operating within a more resilient architecture designed to maintain the integrity of education's fundamental principles and values [24].

As shown in table I below and depicted by a bar graph in figure 5 below, teachers' awareness of various AI tool kinds was also revealed. The most well-known were automated grading tools, adaptive learning systems, and online formative assessments, which 36% and 41% of teachers, respectively, reported being "highly familiar" with. On the other hand, 50% of respondents said they were unfamiliar with the predictive analytics solutions, which was the largest amount. This suggests that even if AI technologies are currently being used in classrooms, most practitioners may not be completely trained to use them or see the need to do so, which is why advanced tools like learning analytics are not widely adopted [25].

TABLE I. FAMILIARITY OF AI TOOLS AMONG EDUCATORS [25].

| AI Tools                           | Familiarity % |        |     |
|------------------------------------|---------------|--------|-----|
|                                    | High          | Medium | Low |
| Intelligent Tutoring Systems (ITS) | 36            | 44     | 20  |
| Chatbots                           | 28            | 47     | 25  |
| AI-Assisted Assessment             | 41            | 38     | 21  |
| Statistical Forecasting            | 19            | 31     | 50  |

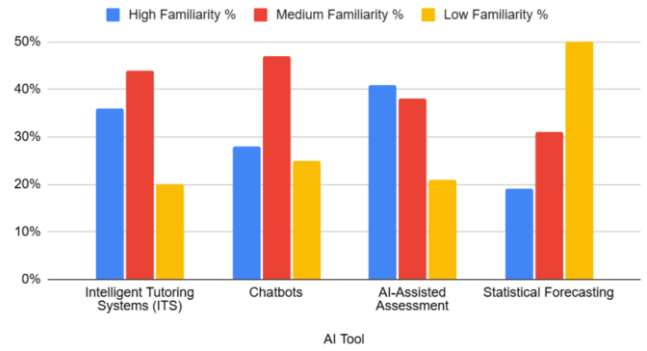


Fig. 5. Survey results indicating varying levels of educator familiarity with AI technologies [25].

To equip educators with AI literacy—the capacity to use AI tools critically, ethically, and effectively, shifts necessitate on significant and ongoing investment in professional development.

#### B. Ensuring Human Oversight and Ethical Governance

A "human-in-the-loop" model is crucial to reducing the risks associated with AI. This method blends the effectiveness of AI with the ethical reasoning and sophisticated judgement of human experts. AI should not completely replace human decision-making; rather, it should support and inform it. Technology platforms need to include teacher override capabilities and be transparent about how AI recommendations are made for this to be effective.

Among their main complaints about online research, participants frequently list a lack of contact channels and a lack of researcher attentiveness. In contrast, individuals are rarely left unattended by researchers in on-site labs. Usually, a team member stays available to address enquiries and resolve any technical issues. The same guidelines should be followed when gathering data online: researchers must keep convenient channels open and be physically present to communicate with participants in a timely and clear manner [26].

The Table II shows how can stakeholders mitigate challenges and avert risks of AI; Students need to indulge in the process of drafting itself, instead of just completing the work at hand. Whereas Educators must maintain their authority over machine, hence AI tools need to be used as assistants rather than replacements. Institutions and Governments should work on data security and better policy to reduce and remove digital divide completely. Contribution from every stakeholder is necessary for a Human-Centric Framework.

There are three main elements that must be present in any AI system to make it trustworthy and reliable. The system should be lawful, ethical and robust. These components are presented by the AI HLEG. A system must comply with all the rules, laws and regulations. Ethical principles need to be at the center of the architecture of the AI model. Lastly to have the ability to adapt to various circumstances and withstand any disruptions, both technical and societal [27].

Ethics in Design mandates that developers and designers take into account the intended use of the AI system they are working on as well as the possible outcomes of its application. This includes taking into account the values (or guiding

TABLE II. ILLUSTRATES HOW STAKEHOLDER CAN MITIGATE CHALLENGES AND AVERT RISKS

| Stakeholder  | Primary Challenge       | Mitigation Strategy                                                                                                                                 |
|--------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Students     | Academic Dishonesty     | “Process over Product” Assessment: Check the drafting process of write-ups and oral defense, rather than weighing on just the final essay.          |
| Educators    | Loss of Agency          | “Human-in-the-loop” Protocols: AI suggests lesson plans, but teachers must explicitly review and modify them as per the students.                   |
| Institutions | Data Privacy & Security | Local/Private LLMs: Use walled-garden AI instances, e.g., University-hosted GPT, that do not train on student data.                                 |
| Governments  | Digital Divide          | Offline-First AI: Prioritize lightweight models (e.g., Gemini Nano, GPT-4o mini, Qwen etc.) that run on basic hardware without high-speed internet. |

principles) that are represented in the AI system's architecture and goals. There are similarities between this and initiatives to integrate human values into software engineering. Anticipating the consequences of disregarding an ethical standard can also draw attention to possible problems that might otherwise go unnoticed [28].

Protecting AI systems from security threats requires a multifaceted strategy that combines technical solutions with policy enforcement. Strict controls must be put in place on the input side to ensure that data is inspected, cleaned, and handled appropriately before being fed into AI models. To make sure that only clean, intended data affects the learning process, methods for identifying or stopping data poisoning attempts are also essential to maintain the privacy.

When humans and AI work together to achieve a common objective, this is known as human-AI collaboration. The goal is to establish synergistic partnerships where AI and humans work together to achieve successful results in a variety of fields. By combining the effective computational abilities of AI in finding patterns, extracting valuable observations from datasets and providing recommendations and predications based on data, with the human expertise of domain, creativity, ethics and cognitive capabilities, we can create a system that can capitalize on their strengths and address complex issues, make well-thought decisions and spur innovation [29].

Institutions need to also implement strict governance guidelines. This includes providing strict data privacy policies based on the principle of ethical use of data into, requiring transparency from owners, and performing bias checks of the models before the actual deployment.

### C. Global Policies to Overcome Misuse of AI

In order to ensure ethical governance of AI, several government national/international, have taken steps. With the groundbreaking AI Act, the European Union (EU) is leading regulatory efforts, tackling the rapidly developing field of artificial intelligence (AI). The AI Act includes third-country providers and deployers who put AI systems on the EU market in addition to the organisations creating and implementing AI inside the EU.

AI security systems can be made more transparent by utilising Explainable AI (XAI) approaches. Finding biases or vulnerabilities in the AI used for other reasons requires this transparency. Threat hunting enabled by AI is also capable of examining large datasets pertaining to the creation and application of AI, spotting any abuse attempts by spotting

questionable trends. But there are certain drawbacks to this strategy. Security AI must constantly adapt due to the dynamic nature of AI threats [30].

The fabrics established by and NIST stress the need of public mindfulness regarding the harms of AI and organizational enterprise in handling AI pitfalls successfully. According to these documents, further awareness may affect in better threat operation procedures and a decline in the abuse of AI. Likewise, the UN report from 2023 emphasizes the need for legislation and public education to help AI abuse, showing that organizations and the general public are more suitable to identify and reduce hazards related to AI when they're informed [30].

A report from the EU Parliament in 2019 proposed several policies for regulating algorithms after looking at all the related social and technical issues. The solutions have four parts. First is to make common people more aware about the use of AI, through education and supporting watchdogs. Secondly, make government agencies accountable for their use of AI. Third part involved legal oversight and implementing laws, at last, to make sure that everyone and every government around the world cooperate with these rules [31].

The IEEE is developing an ethics standard, P7003, which is mainly focusing on algorithmic bias. It gives developers a set of guidelines to prevent their systems from having outcomes that are unintended, unjustified, or unfairly discriminatory [31].

This initiative from IEEE is of a high value as it sets clear instructions for developers and programmers as to how their models should behave like. It will prevent biasness and improve the user experience of AI models.

Only 4.5% of laws proposed by government bodies, suggest more stringent regulation, while the great majority (98%) only function as "soft laws," that is, rules that do not impose any kind of legal responsibility. Canada, Germany, and the United Kingdom are leading this still weak trend, with the United States, Australia, and Norway following closely behind [32].

Countries like Canada, Germany, and the United Kingdom appear to be increasingly acknowledging that "ethics" and "compliance" are two different areas. The number of legislative records on AI-related bills increased from one in 2016 to eighteen in 2021, with the United States, Spain, and

the United Kingdom ranking as the top three "proto-AI-legislators" of 2021. This suggests that we may be entering a transitional period where these ideas may soon be turned into laws [32].

## V. CONCLUSION

Artificial Intelligence is a tool that has duality for education; a technology of immense power and a significant drawback. The implicit to epitomize literacy, enhance educator's effectiveness, and expand availability offers a compelling vision for a more effective and indifferent educational future. Still, this vision is shadowed by hazardous challenges, including algorithmic bias, data sequestration pitfalls, a widening digital peak, and the implicit dehumanization of the literacy process.

The way forward is one of responsible and deliberate integration rather than blind acceptance or frightful rejection. This calls for a human-centred approach that reinterprets the teacher's function as a deep learning facilitator, makes significant investments in professional growth, and creates strong ethical and policy boundaries. The human decisions that direct the application of AI will ultimately determine its impact, not the complexity of its algorithms. AI can be used as a potent instrument to enhance human capability and realise each learner's full potential by putting equity, safety, and the priceless value of human connection first.

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